

Workshop

Biostatistics & SPSS Workshop

Clear concepts • correct test choice • practical analysis

Presented by
Safoorah Sabir
Senior Lecturer Statistics



Why this workshop matters

1

From data to answer

Convert patient records, forms and scores into clear summaries and decisions.

3

Use SPSS with confidence

Enter data correctly, select menus and read key output.

2

Avoid wrong tests

Choose the correct analysis according to variable type and study question.

4

Report simply

Write results in a clean format that readers can understand.



Main idea: statistics is not only calculation — it is a way to answer clinical questions responsibly.

Workshop roadmap

1 Variables & data types

What to know, when to use it.

2 SPSS data setup

What to know, when to use it.

3 Descriptive statistics

What to know, when to use it.

4 Sampling & inference

What to know, when to use it.

5 Hypothesis testing

What to know, when to use it.

6 Common statistical tests

What to know, when to use it.

7 Reporting & practice

What to know, when to use it.

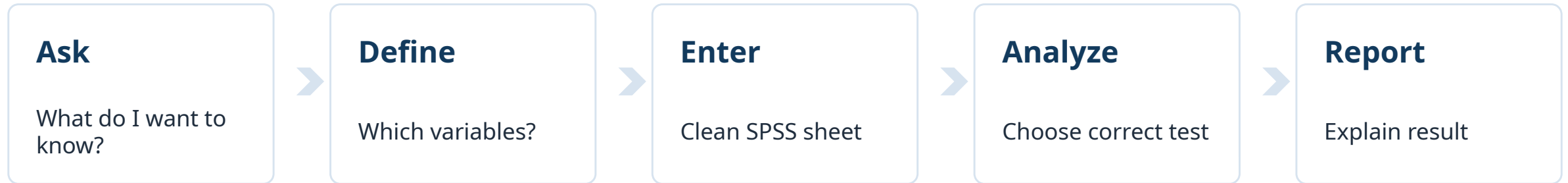
The order moves from “What is my data?” to “Which test should I apply?”

Learning outcomes

- Identify qualitative, quantitative, continuous and discrete variables.
- Set variable names, labels, values and measurement scale in SPSS.
- Summarize data using mean, median, mode, variance and standard deviation.
- Understand sampling, inference and hypothesis testing in simple words.
- Choose between t-tests, ANOVA, chi-square and regression for common questions.



From a question to an answer

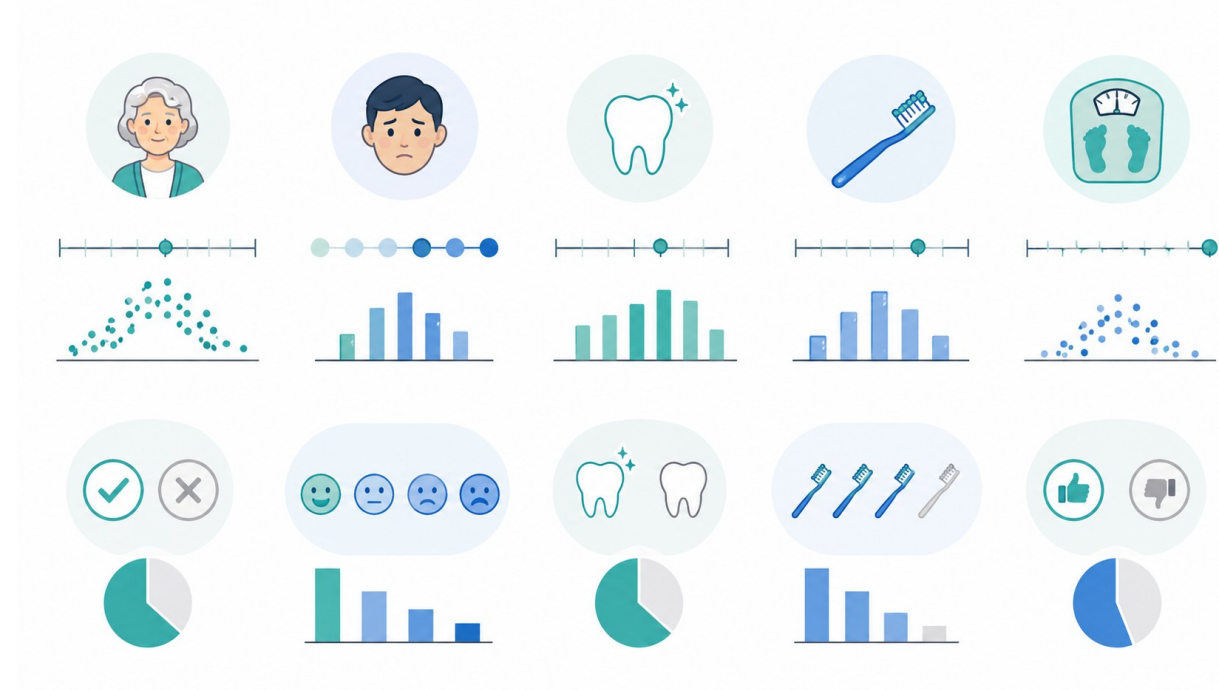


Example question: Is the average pain score lower after treatment than before treatment?

Part 1

Biostatistics basics

Understand the language of variables before opening SPSS.



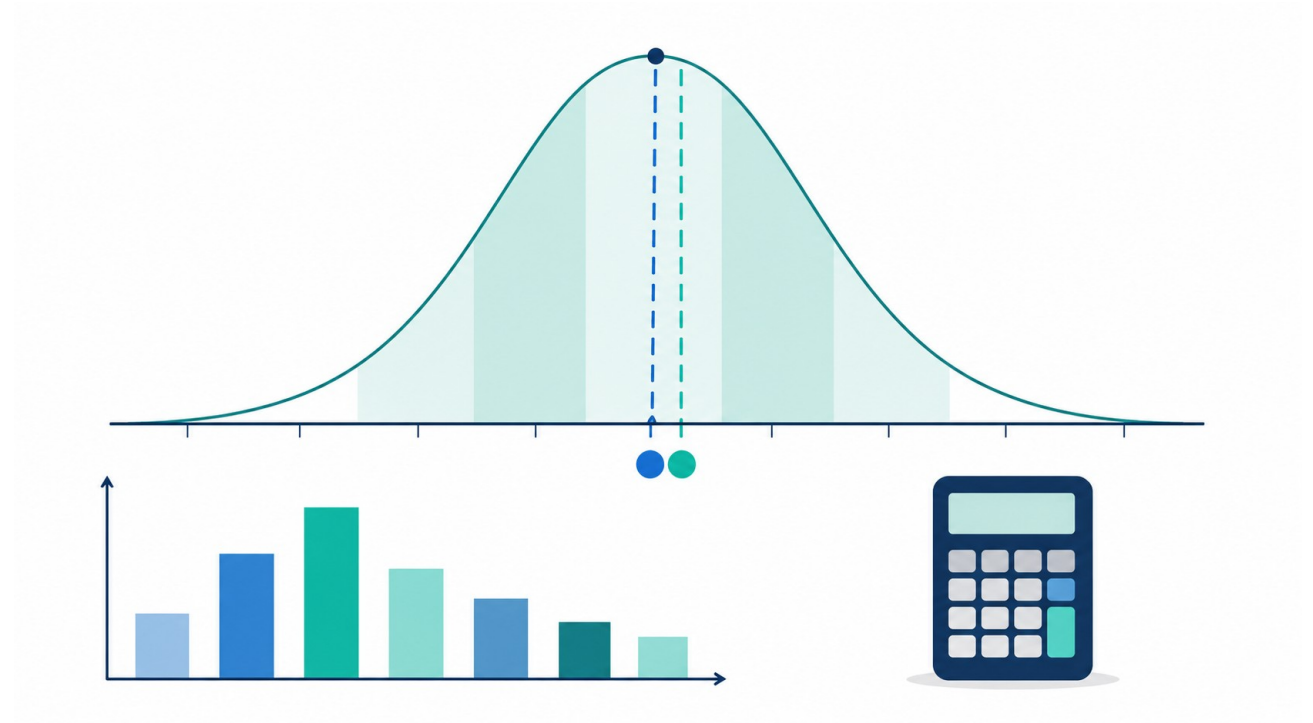
What is biostatistics?

✓ Simple definition

Biostatistics uses numbers, summaries and tests to answer health-related questions from data.

→ In practice

It helps us describe patients, compare groups and study relationships between variables.



Qualitative vs quantitative variables

A Qualitative variables

Categories or names.

Examples: gender, blood group, brushing habit, satisfaction level.

Quantitative variables

Numbers that can be measured or counted.

Examples: age, pain score, mouth opening in mm, appointment duration.



Continuous vs discrete variables

Continuous

Measured on a scale and can have decimals.
Examples: age, weight, time, pocket depth in mm.

Discrete

Counted as whole numbers.
Examples: number of visits, number of missing teeth, number of medicines.

Quick rule: measured = continuous; counted = discrete

Dependent vs independent variables

X Independent variable

The factor or group that may explain a change.
Examples: treatment type, age group, brushing frequency.



Y Dependent variable

The outcome we measure.
Examples: pain score, plaque score, healing time, satisfaction.

Example: Does mouthwash type change plaque score?

Mouthwash type = independent variable | Plaque score = dependent variable

Data types in SPSS

1 Numeric

Numbers used for analysis: age, pain score, visits.

T String

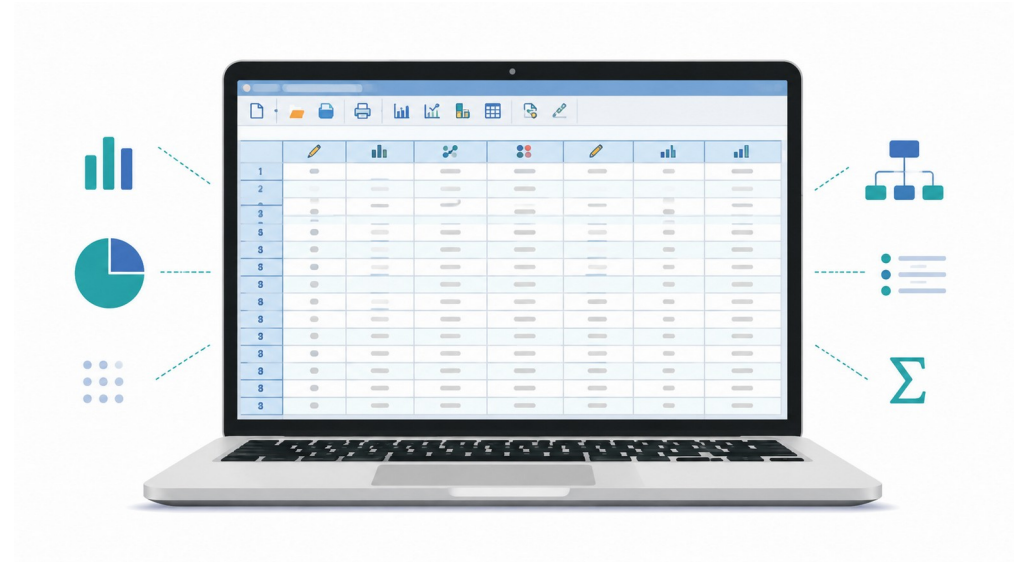
Text or labels: name, city, remarks. Not ideal for tests.

D Date

Dates of visit, procedure or follow-up.

\$ Comma / Dollar

Special formats; usually not needed for basic analysis.



Tip: use numeric coding for categories whenever possible.

SPSS Variable View: key columns

Question	Groups / Data	Outcome	Test	
Name	Short variable name: age, gender, pain_pre			
Type	Numeric, string, date			
Label	Full explanation of the variable			
Values	Coding for categories: 1 = Male, 2 = Female			
Measure	Nominal, ordinal or scale			

Good setup makes analysis easier and reduces mistakes.

Measurement scale: nominal, ordinal, scale

N Nominal

Categories with no order.
Examples: gender, treatment group, yes/no.

O Ordinal

Categories with order.
Examples: mild / moderate / severe, satisfaction level.

S Scale

Numerical measurement.
Examples: age, pain score, plaque score, time in minutes.

SPSS uses measurement scale to guide summaries and graphs.

How to choose the correct measurement scale

Question	Groups / Data	Outcome	Test	
If data are names only	Male/Female, Yes/No, Group A/B	Nominal	Frequencies, chi-square	
If data have order	Mild/Moderate/Severe	Ordinal	Median, percentages	
If data are measured numbers	Age, pain score, duration	Scale	Mean, SD, t-test, ANOVA	

Simple memory: Name = Nominal | Order = Ordinal | Number = Scale

Organizing data correctly



One row = one participant

Each participant should appear once in the main data sheet.



One column = one variable

Do not mix two variables in one column.



Use simple codes

Example: 1 = Yes, 0 = No. Keep a codebook.



Avoid empty text

Use consistent coding for missing data.



Check before analysis

Run frequencies and descriptives first.

Part 2

Describe the data

Start with simple summaries before doing statistical tests.



Descriptive statistics: first look at the data

- Tells us what the data look like.
- Finds the center of values: mean, median, mode.
- Finds spread: variance and standard deviation.
- Identifies unusual values and skewness.



Mean, median and mode

M Mean

Average value.
Example: average pain score
after treatment.

Md Median

Middle value after sorting.
Useful when data are skewed.

Mo Mode

Most repeated value.
Useful for categories and
common responses.

Example values: 2, 3, 3, 4, 8 → mean = 4, median = 3, mode = 3

Variance and standard deviation

V Variance

Average squared distance from the mean. It tells how scattered the values are.

SD Standard deviation

Average spread around the mean in the original unit.

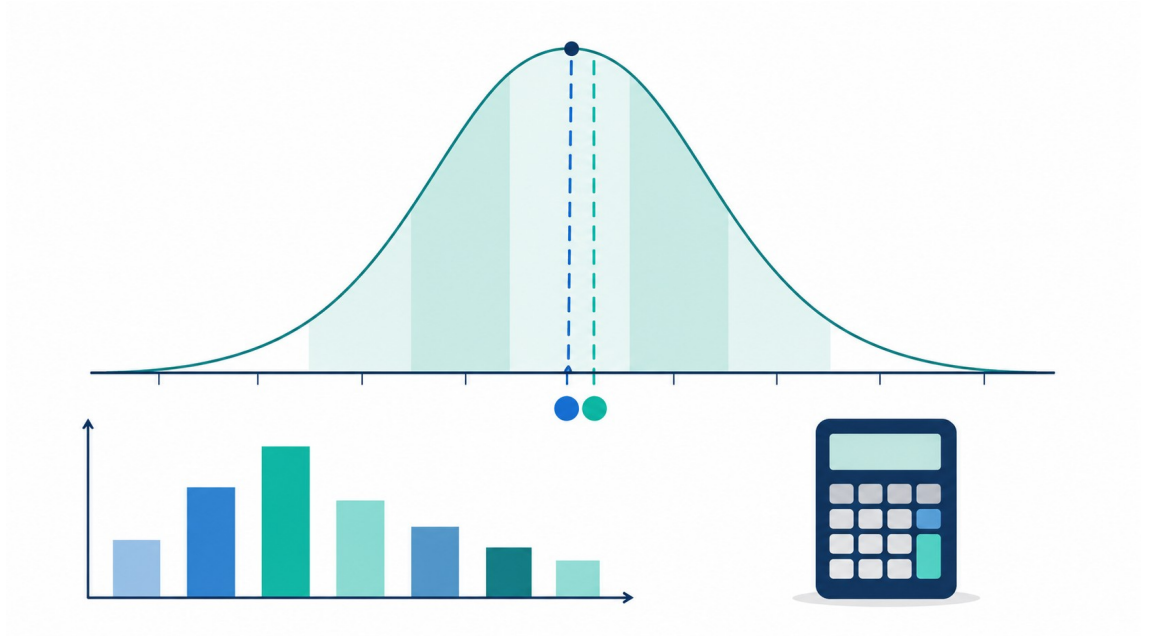
! Easy meaning

Small SD = values are close.
Large SD = values are spread out.

Example: two groups may have the same mean, but different spread.

Symmetric distribution

- Values are balanced around the center.
- Mean, median and mode are close to each other.
- Many parametric tests work better when data are roughly symmetric.
- Always check histogram or normality before deciding.



Skewness: when data are not balanced

+ Positive skew

Most values are low, but a few values are very high.
Example: long treatment time for a few complex cases.

- Negative skew

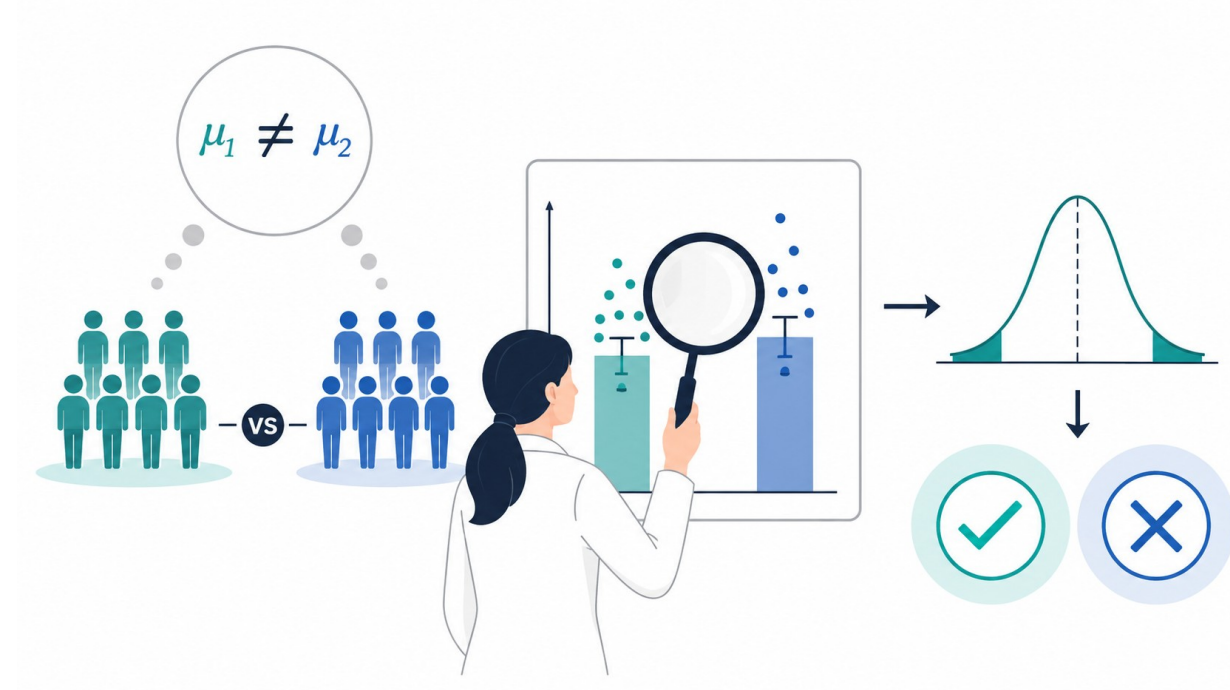
Most values are high, but a few values are very low.
Example: high satisfaction with a few low scores.

For skewed data, median can be more meaningful than mean.

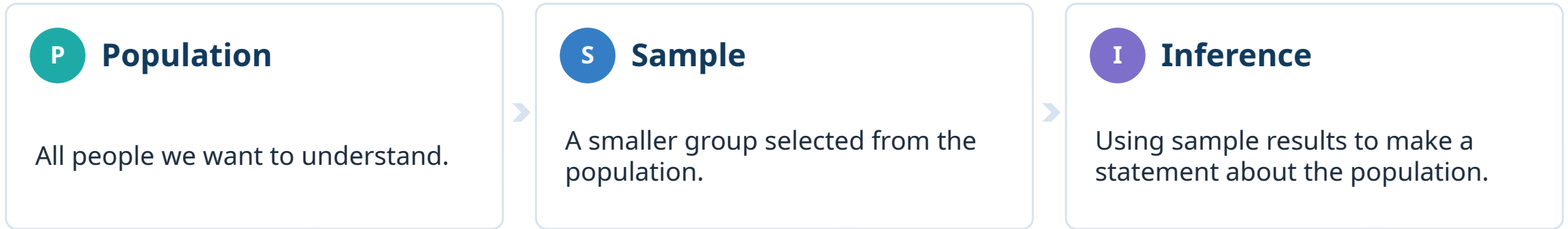
Part 3

Inference & testing

Move from sample summaries to evidence-based conclusions.



Sampling: why we do not study everyone



Good sample = better confidence in your result.

Inference in simple words

1

Collect data from a sample.

2

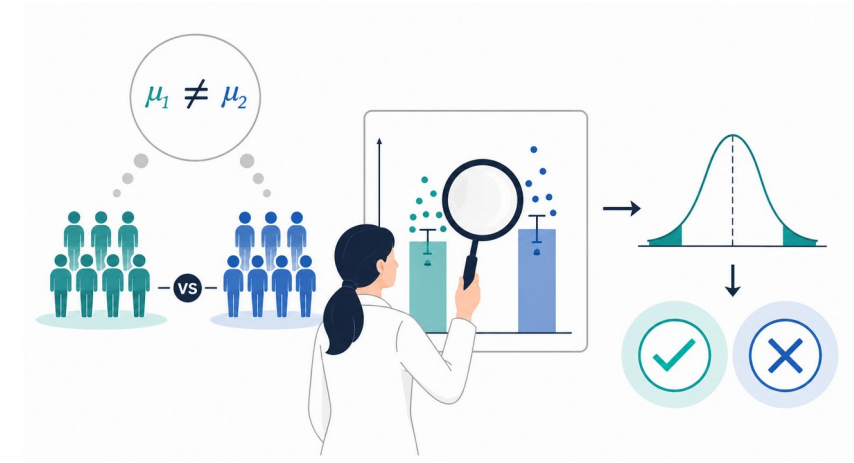
Calculate a statistic: mean, percentage or difference.

3

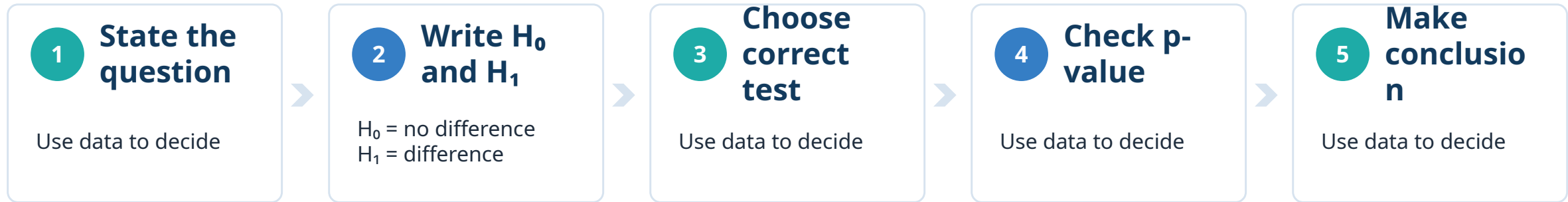
Estimate what may be true in the larger group.

4

Use a test to judge if the finding is likely real or due to chance.



Hypothesis testing: easy steps



Hypothesis testing does not prove; it gives evidence.

P-value and significance

P-value

Probability of getting a result like this if the null hypothesis is true.

Common rule

If $p < 0.05$, the result is called statistically significant.

Important: statistical significance is not always clinical importance.

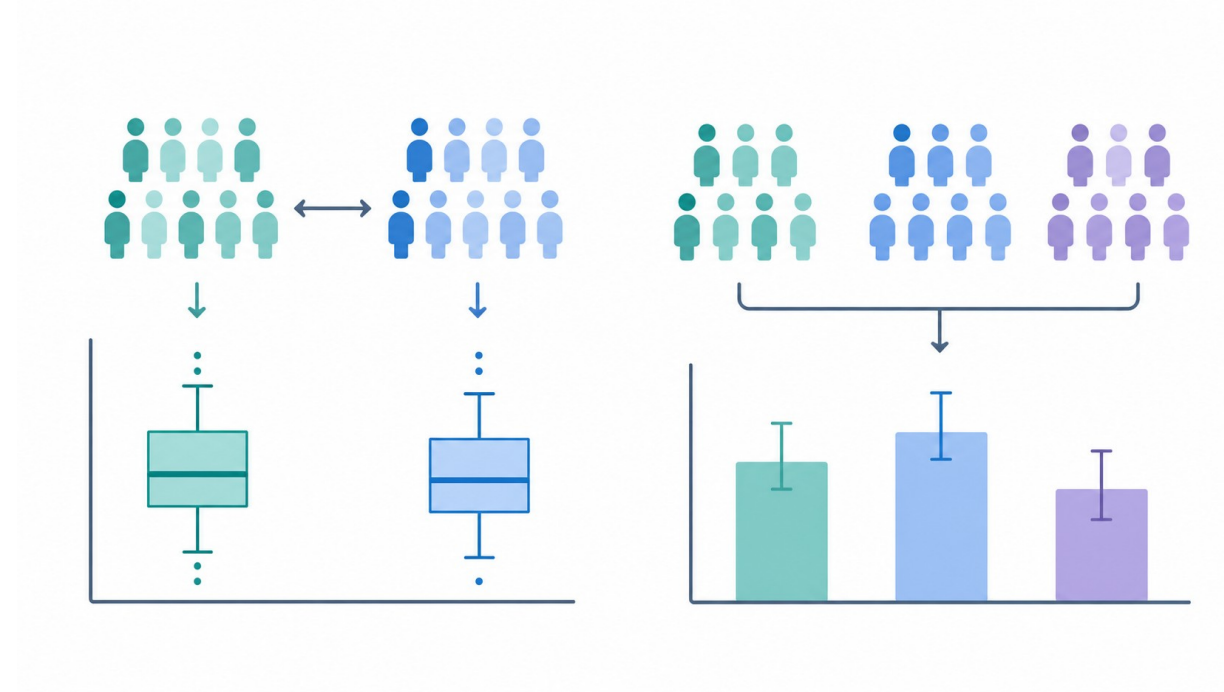
Which test should I use?

Question	Groups / Data	Outcome	Test
One mean vs fixed value	1 group	Scale	One-sample t-test
Two separate groups	2 independent groups	Scale	Independent t-test
Same participants before/after	2 related readings	Scale	Paired t-test
Three or more groups	3+ independent groups	Scale	ANOVA
Association of categories	Two categorical variables	Nominal/Ordinal	Chi-square
Prediction/relationship	Outcome + predictor	Mostly scale	Regression

Part 4

Common tests

Learn the question, data type, menu route and simple interpretation.



One-sample t-test

? Question

Is one sample mean different from a known value?

Ex Example

Is average pain score different from 5?

S SPSS menu

Analyze → Compare Means → One-Sample T Test

Data needed: one scale outcome variable.

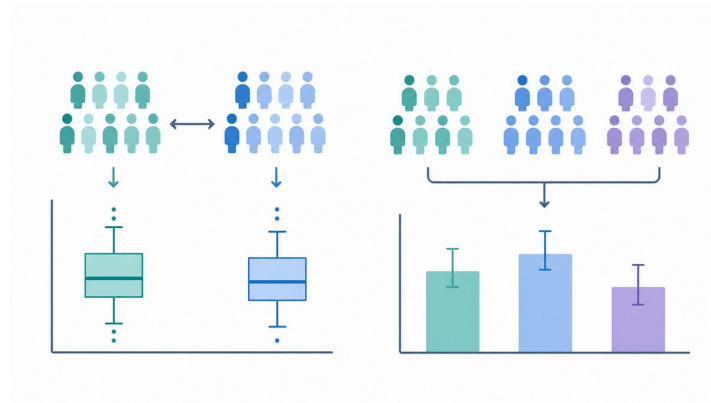
Independent-samples t-test

? Question

Do two separate groups have different mean outcome values?

Ex Example

Compare mean plaque score between two treatment groups.



SPSS: Analyze → Compare Means → Independent-Samples T Test

Paired-samples t-test

? Question

Does the mean change when the same participants are measured twice?

Ex Example

Pain score before treatment vs pain score after treatment in the same participants.

Key idea: two readings are connected, not independent.

SPSS: Analyze → Compare Means → Paired-Samples T Test

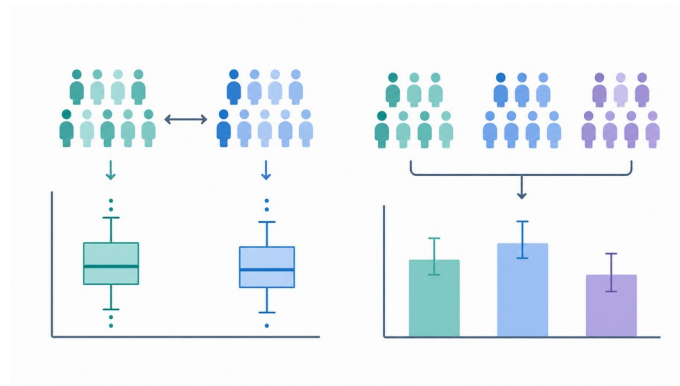
ANOVA

? Question

Do three or more independent groups have different mean values?

Ex Example

Compare mean plaque score among three toothpaste groups.



If ANOVA is significant, use post-hoc tests to see which groups differ.

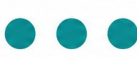
Chi-square test

? Question

Are two categorical variables associated?

Ex Example

Is treatment group associated with improvement: Yes/No?

SPSS: Analyze → Descriptive Statistics → Crosstabs → Chi-square

Regression

? Question

Can one or more variables explain or predict an outcome?

Ex Example

Can age, brushing frequency and baseline score predict final plaque score?



How to report results simply

1 Describe

Report mean $\pm$ SD or frequency (%).

2 Test

Mention the statistical test used.

3 Conclude

Write p-value and plain-language meaning.

Example: Mean pain score decreased from 6.1 ± 1.4 to 3.2 ± 1.1 after treatment (paired t-test, $p < 0.001$).

Common mistakes to avoid in SPSS

- Entering categories as text when numeric codes are easier.
- Choosing “Scale” for categorical data such as Yes/No.
- Using independent t-test for before/after readings.
- Ignoring missing values and outliers before analysis.
- Reporting p-value without explaining what it means.

Hands-on activity

1 Step 1

Create variables in Variable View.

2 Step 2

Enter 10 sample records in Data View.

3 Step 3

Run frequencies and descriptives.

4 Step 4

Choose a test using the decision table.

5 Step 5

Write a one-line interpretation.

Key take-home messages

- Start with the research question, not the software menu.
- Identify variable type before choosing statistics.
- Use descriptive statistics before inferential tests.
- Choose tests according to groups, outcome type and data structure.
- Report results with both numbers and plain meaning.

More study & reference material



Kent State University Libraries: SPSS Recode Variables

Step-by-step guide for recoding old values into new categories in SPSS.

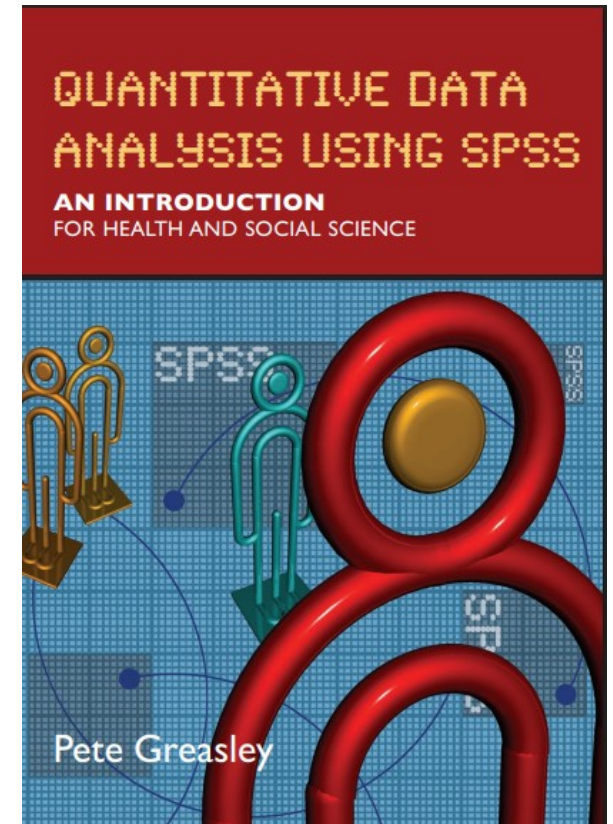
<https://libguides.library.kent.edu/SPSS/RecodeVariables>



Recommended book

Quantitative Data Analysis Using SPSS: An Introduction for Health and Social Science — Pete Greasley

Use these resources after the workshop to repeat SPSS commands, practice recoding, and revise basic analysis steps.



Beginner-friendly SPSS reading for basic data analysis practice.



Thank you

Questions & Discussion

Safoorah Sabir